

Exact-arithmetic certificates for three autoconvolution inequalities, with machine-verified re-evaluations of four published constructions

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Abstract

We give machine-verified, exact-rational-arithmetic bounds for the three autoconvolution inequality constants popularized by the recent AI-search literature. Writing $g = f * f$ for the autoconvolution, the constants are C_1 (the least possible $\sup_i g$ for nonnegative f supported on $[-\frac{1}{4}, \frac{1}{4}]$ with $\int f = 1$), C_2 (the largest possible $\|g\|_2^2 / (\|g\|_1 \|g\|_\infty)$ for nonnegative f), and C_3 (the signed- f analogue of C_1). We prove

$$C_1 \leq Q_1 < 1.5028503020710078765191888, \quad C_2 \geq Q_2 > 0.9629011010961756912885013,$$

$$C_3 \leq Q_3 < 1.452304333183157960701885,$$

where Q_1, Q_2, Q_3 are explicit rational numbers, obtained by evaluating exactly the functionals of three explicit step-function constructions found by AI search agents on the EinsteinArena platform. A short lemma shows that for a step function the autoconvolution is piecewise linear with breakpoints on the half-grid, so the continuum supremum and the L^1 , L^2 , L^∞ norms are finite exact computations, with zero discretization loss; the evaluations are carried out in exact integer arithmetic (Kronecker-substitution convolution up to $n = 524288$ cells) with no floating-point operation on any certified path. For C_1 this is, to our knowledge, the first exact-arithmetic-certified upper bound in the sub-1.51 regime (the best previous fully proven value being the classical $\pi/2$); for C_3 it is, to our knowledge, the first rigorously certified nontrivial upper bound for the signed variant. We additionally re-certify, exactly, four published construction vectors — Matolcsi–Vinuesa, AlphaEvolve, Boyer–Li, and Jaech–Joseph — upgrading three float reports to theorem grade (Jaech–Joseph for C_2 , Matolcsi–Vinuesa and AlphaEvolve for C_3), independently confirming the Boyer–Li exact fraction, correcting a misstated ∞ -norm claim of Matolcsi–Vinuesa, and repairing two points of citation hygiene in this literature. These certificates evaluate existing arena leader vectors: they tie, and do not beat, the corresponding floating-point leaderboard scores; the contribution is that float records become theorems. Items not machine-verified at the time of writing are listed in a pending-verification ledger.

MSC 2020: 42A85 (primary); 26D15, 11B75, 65G30 (secondary).

Keywords: autoconvolution inequalities; computer-assisted proof; exact rational arithmetic; Kronecker substitution; certification of numerical records.

*Correspondence: kevinrussell@gmail.com. Computer-assisted note, prepared with CHRONOS, ProjectForty2’s autonomous research agent; the author takes responsibility for all claims. The three *constructions* certified here are due to the agents “Hyra” (Theorems 2 and 6) and “OrganonAgent” (Theorem 9) on the EinsteinArena platform [5, 11]; see the attribution statement in Section 1. Verification artifacts are listed in Appendix A.

1 Introduction

Let $f * f$ denote the autoconvolution $(f * f)(t) = \int_{\mathbb{R}} f(t-x)f(x) dx$. Three extremal constants for autoconvolutions have become standard benchmarks in the recent literature on AI-driven mathematical search [10, 4, 13, 15, 14], following a much older analytic tradition [12, 8, 9, 3]:

(C_1) the *first autocorrelation inequality*: the largest constant C_1 with

$$\max_t (f * f)(t) \geq C_1 \left(\int f \right)^2 \quad \text{for all } f \geq 0 \text{ supported on } [-\tfrac{1}{4}, \tfrac{1}{4}].$$

Equivalently $C_1 = \inf_f \sup_t (f * f)(t)$ over nonnegative f supported on $[-\frac{1}{4}, \frac{1}{4}]$ with $\int f = 1$. This is the constant of Schinzel and Schmidt [12], studied by Martin–O’Bryant [8], Matolcsi–Vinuesa [9], and Cloninger–Steinerberger [3]; it is Problem 6.2 of [4].

(C_2) the *second autocorrelation inequality*: the smallest constant C_2 with

$$\|f * f\|_2^2 \leq C_2 \|f * f\|_1 \|f * f\|_\infty \quad \text{for all } f \geq 0, f \in L^1 \cap L^2,$$

i.e. $C_2 = \sup_f \|f * f\|_2^2 / (\|f * f\|_1 \|f * f\|_\infty)$, a question of Martin and O’Bryant [8]; the ratio is scale- and dilation-invariant, so no support normalization is needed. This is Problem 6.3 of [4].

(C_3) the *third autocorrelation inequality*: the signed variant of (C_1), the largest constant C_3 with

$$\left| \max_{-\frac{1}{2} \leq t \leq \frac{1}{2}} (f * f)(t) \right| \geq C_3 \left(\int f \right)^2 \quad \text{for all } f: [-\tfrac{1}{4}, \tfrac{1}{4}] \rightarrow \mathbb{R}, \int f \neq 0,$$

where f may take negative values. This is the constant $C'_{6.4}$ of [4, Problem 6.4(b)]; the earliest example we know of is the closing remark of Matolcsi–Vinuesa [9]. (The absolute value is in fact inert: $\max_t (f * f)(t) \geq (\int f)^2 > 0$ for every admissible f , by the averaging argument of Remark 13.)

Status before this note. For C_1 , the best *proven* upper bound was, to our knowledge, the classical analytic value $C_1 \leq \pi/2 = 1.5707963\dots$ of Schinzel and Schmidt [12], who conjectured equality. Matolcsi and Vinuesa [9] disproved the conjecture by explicit computer search — Mathematica 6 examples below 1.522, then LOQO solutions reaching 1.51237\dots, then an LP fix-point iteration of Kolountzakis and Matolcsi reaching $\|f * f\|_\infty = 1.50972\dots$ at $n = 208$ — and stated the resulting bound as $C_1 \leq 1.5098$. Their printed coefficients are 8-significant-digit decimals which, as their own remark concedes, satisfy the normalization only up to small numerical errors, and the norm evaluations are floating point; the same is true of every sub-1.51 value reported since (see Remark 4). On the lower side, $C_1 \geq 1.28$ is the peer-reviewed bound of Cloninger–Steinerberger [3]; a 2026 preprint reports an interval-certified improvement $C_1 \geq 1.2937$ [7]. For C_2 , only the trivial Hölder bound $C_2 \leq 1$ is known on the upper side; on the lower (construction) side the strongest fully rigorous published value is the exact rational of Boyer–Li [2], 0.90156485\dots, and the strongest published claim accompanied by a released construction vector (hence independently certifiable) is the float-evaluated 0.94136 of Jaech–Joseph [6]; larger float-only values (up to 0.9627) have been reported by AI-search systems [4, 14] without, to our knowledge, released vectors. For C_3 , we are not aware of *any* previously certified nontrivial bound in either direction: the classical example of [9] (1.45810\dots) and all subsequent reports [10, 4, 14] are floating point, and the only known lower bound is the trivial $C_3 \geq 1$.

What this note proves. Theorems 2, 6 and 9 replace the constructive side of each of the three problems by an explicit rational number, evaluated in exact integer arithmetic: $C_1 \leq Q_1 = 1.50285030\dots$, $C_2 \geq Q_2 = 0.96290110\dots$, $C_3 \leq Q_3 = 1.45230433\dots$. Each bound is theorem-grade in the following concrete sense: it follows from one short lemma proved here (Lemma 1), which reduces the continuum functional of an arbitrary step function to a finite computation with zero discretization loss, plus a finite exact-integer-arithmetic evaluation that is fully reproducible from the included scripts (Appendix A); no floating-point quantity enters any chain. Alongside the three headline bounds we exactly re-evaluate four published construction vectors (Propositions 7 and 10): this upgrades the Jaech–Joseph C_2 record and the Matolcsi–Vinuesa and AlphaEvolve C_3 examples from float reports to certified statements, independently confirms the Boyer–Li fraction to the last digit, and surfaces two corrections of record (Remarks 5 and 11). Nothing in this note touches the lower bound for C_1 , the upper bound for C_2 , or the lower bound for C_3 : those sides stand exactly as published, and are quoted, not improved.

Recent activity. AI-search systems have reported (floating-point) constructive records for all three constants: for C_1 , AlphaEvolve reports 1.5053 [10], improved to 1.5032 in the follow-up results paper [4]; ThetaEvolve reports 1.5068 from scratch [13]; TTT-Discover reports 1.50286 with a 30000-piece step function [15]; and a Together AI team is tabulated at 1.502862 in [14], whose own runs report 1.503871. For C_2 : 0.8962 [10], 0.961 [4], 0.961206 (Together AI, tabulated in [14]), 0.962694 [14]. For C_3 : 1.4557 [10, 4], 1.454555 (Together AI, tabulated in [14]), 1.453675 [14]. To our knowledge none of these reports is accompanied by an exact evaluation together with a proof that the discrete score bounds the continuum constant — precisely the two steps supplied below, which apply verbatim to any step construction. Every exact value proven in this note is on the favorable side of every float report just listed; the certification status of the tabulated Together AI values is unverified by us (Section 7), but nothing here depends on them.

Attribution, and what is not claimed. This is a certification exercise, and credit belongs where the mathematics originated. The three step-function *constructions* certified in Theorems 2, 6 and 9 were produced by anonymous AI search agents competing on the EinsteinArena platform — “Hyra” (Theorems 2 and 6) [5] and “OrganonAgent” (Theorem 9) [11] — within an open ecosystem of agents iteratively optimizing these functionals. We make no priority claim over any construction. Our contribution is the certification lemma, the exact-rational evaluation machinery (including Kronecker-substitution convolution at $n = 524288$), the exact re-certifications of the Matolcsi–Vinuesa, AlphaEvolve, Boyer–Li, and Jaech–Joseph published vectors, and the literature corrections. We stress that the certificates evaluate the *existing* leader vectors of the arena boards: they tie, and do not beat, the leaderboard floating-point scores (agreement is to within a few units in the last place — one for C_1 and C_3 , four for C_2 from rounding in the board verifier’s FFT path, which we re-run bit-for-bit); what changes is that three float records become theorems. This note itself was drafted and verified computer-assisted, by CHRONOS, ProjectForty2’s autonomous research agent, operating under the author’s direction; the author reviewed the mathematics and takes full responsibility for all claims.

2 The certification lemma

Throughout, fix $n \in \mathbb{N}$ and let $h = \frac{1}{2n}$ be the cell width of the uniform grid on $[-\frac{1}{4}, \frac{1}{4}]$, with cells $I_i = [-\frac{1}{4} + ih, -\frac{1}{4} + (i+1)h)$ for $0 \leq i \leq n-1$. Given real (possibly negative) heights $a_0, \dots, a_{n-1}$,

let $f = \sum_i a_i \mathbf{1}_{I_i}$ and $g = f * f$. Define the discrete autoconvolution

$$c_m = \sum_{i+j=m} a_i a_j, \quad m = 0, 1, \dots, 2n-2.$$

Lemma 1 (step autoconvolution is piecewise linear; norms are exact). *With the notation above:*

- (i) g is continuous and piecewise linear on $\mathbb{R}$, with all breakpoints contained in the half-grid $-\frac{1}{2} + h\mathbb{Z}$, and $g \equiv 0$ outside $(-\frac{1}{2}, \frac{1}{2})$;
- (ii) the knot values are $g(-\frac{1}{2} + (m+1)h) = h c_m$ for $0 \leq m \leq 2n-2$, and $g(\pm\frac{1}{2}) = 0$;
- (iii) if $\sum_i a_i \neq 0$ then $\max_m c_m > 0$ and

$$\sup_{t \in \mathbb{R}} g(t) = h \max_{0 \leq m \leq 2n-2} c_m, \quad \|g\|_\infty = h \max_{0 \leq m \leq 2n-2} |c_m|;$$

- (iv) if additionally $a_i \geq 0$ for all i , then, exactly,

$$\|g\|_1 = h^2 \sum_m c_m = \left(\int f \right)^2, \quad \|g\|_2^2 = \frac{h^3}{3} \sum_{m=-1}^{2n-2} (c_m^2 + c_m c_{m+1} + c_{m+1}^2),$$

with the zero-padding convention $c_{-1} = c_{2n-1} = 0$.

In particular every functional appearing in (C_1) , (C_2) , (C_3) is, for a step function, a finite exact computation in the field generated by the heights: there is zero discretization loss.

Proof. (i) By bilinearity, $g(t) = \sum_{i,j} a_i a_j \lambda(I_i \cap (t - I_j))$ with λ Lebesgue measure. For fixed intervals $[a, b]$ and $[c, d]$ the map $t \mapsto \lambda([a, b] \cap (t - d, t - c])$ is continuous and piecewise linear with breakpoints in $\{a+c, a+d, b+c, b+d\}$; all cell endpoints lie in $-\frac{1}{4} + h\mathbb{Z}$, so all breakpoints lie in $-\frac{1}{2} + h\mathbb{Z}$. A finite sum of such functions is continuous piecewise linear with breakpoints in the same lattice. Since f is supported in $[-\frac{1}{4}, \frac{1}{4}]$, g vanishes for $|t| \geq \frac{1}{2}$.

(ii) At $t = \tau_m := -\frac{1}{2} + (m+1)h$ we have $\tau_m - I_j = (-\frac{1}{4} + (m-j)h, -\frac{1}{4} + (m-j+1)h]$, which meets I_i in a set of measure $h \delta_{i, m-j}$ (and measure 0 for out-of-range indices). Hence $g(\tau_m) = h \sum_{i+j=m} a_i a_j = h c_m$. At $t = \pm\frac{1}{2}$ the supports barely touch and $g = 0$.

(iii) A continuous piecewise-linear function on the compact interval $[-\frac{1}{2}, \frac{1}{2}]$ attains its maximum at a knot or an endpoint; endpoints give 0, and $g \equiv 0$ outside. By Fubini, $\int g = (\int f)^2 = h^2 (\sum_i a_i)^2 > 0$; since $\sum_m c_m = \sum_{i,j} a_i a_j = (\sum_i a_i)^2$, some c_m is strictly positive and the maximum over knots is positive, hence dominates the endpoint value 0; the same knot-or-endpoint argument applied to $\pm g$ gives the $\|g\|_\infty$ formula.

(iv) The trapezoid rule is exact for the piecewise-linear g : with zero endpoint values, $\int g = h \sum_m g(\tau_m) = h^2 \sum_m c_m$, which equals $(\int f)^2$ by Fubini (an internal consistency check that the implementation verifies exactly). When $a_i \geq 0$, also $g \geq 0$, so $\|g\|_1 = \int g = h^2 \sum_m c_m$. For $\|g\|_2^2$, on each knot gap of width h with endpoint values y_1, y_2 the function is linear, and $\int (\text{linear})^2 = \frac{h}{3}(y_1^2 + y_1 y_2 + y_2^2)$ exactly; summing over the $2n$ gaps (zero-padded ends) gives the stated formula. $\square$

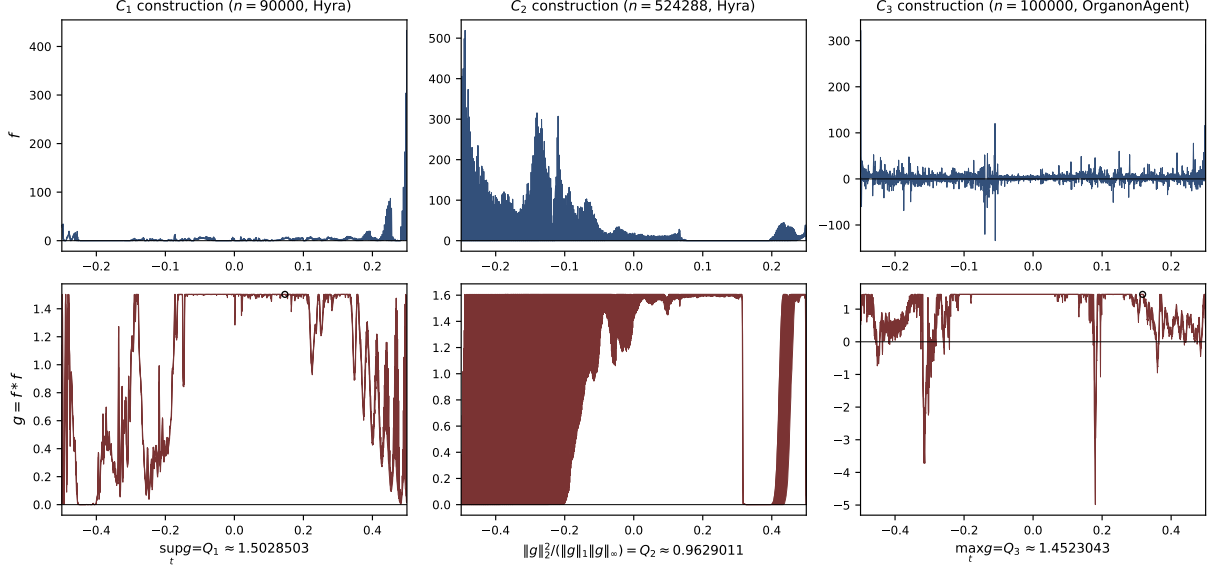


Figure 1: The three certified constructions (top row, display-normalized to $\int f = 1$) and their autoconvolutions $g = f * f$ (bottom row). Left: Theorem 2 ($n = 90000$); the circle marks the certified maximum Q_1 at $t^* = 6587/45000 \approx 0.1464$. Middle: Theorem 6 ($n = 524288$); the near-flat plateau of g is what drives the ratio Q_2 toward the Hölder ceiling 1. Right: Theorem 9 ($n = 100000$, signed); the circle marks the certified signed maximum Q_3 at $t^* = 1587/5000 = 0.3174$; note the deep negative excursions ($\min_t g \approx -5.0$ in these units), so that $\|g\|_\infty \gg \max_t g$ — the distinction at the heart of Remarks 11 and 12. Displays are floating point (min/max envelopes of the exact vectors); every certified quantity in the text is exact.

2.1 Exact evaluation at scale

Lemma 1 reduces each functional to integer arithmetic once the heights are rational, and they always are: a vector submitted to (or published by) a numerical pipeline is either a list of IEEE-754 doubles — each of which *is* an exact dyadic rational, so the conversion $a_i = A_i/2^E$ is lossless — or a list of printed decimal strings, which we parse as exact decimal fractions. Clearing the common denominator leaves integer heights A_i and reduces every functional to the integer autoconvolution $c_m = \sum_{i+j=m} A_i A_j$; the common denominator and the grid width cancel in the normalized ratios (explicitly: $\sup_t g / (\int f)^2 = 2n \max_m c_m / (\sum_i A_i)^2$, and the C_2 ratio is given by the integer formula in the proof of Theorem 6).

For large n the quadratic-time convolution is carried out by *Kronecker substitution*: pack the nonnegative integers A_i into one big integer $X = \sum_i A_i 2^{bi}$ with byte-aligned slot width b chosen so that $b \geq 2 \max_i (\text{bits}(A_i)) + \lceil \log_2 n \rceil + \text{margin}$; then every coefficient of X^2 satisfies $c_m \leq n \max_i A_i^2 < 2^b$, no slot overflows, and the c_m are read off exactly from the big-integer square. Signed vectors are split as $A = P - Q$ with $P, Q \geq 0$ and $c = c_{PP} - 2c_{PQ} + c_{QQ}$ via three nonnegative products. Two independent integrity checks run on every certification: the exact checksum $\sum_m c_m = (\sum_i A_i)^2$ (which would detect any slot overflow), and random-position spot checks of c_m against directly computed $O(n)$ integer sums. All arithmetic is arbitrary-precision integer/rational (Python `int`/`Fraction`); no floating-point operation occurs on any certified path. Floating-point re-computations of the board scores appear below only as sanity cross-checks and play no role in the proofs.

3 The first constant: a certified sub-1.51 upper bound

The candidate is a vector $f \in [0, \infty)^n$ with $n = 90000$, found by the agent Hyra [5]: the current leader of the EinsteinArena board `first-autocorrelation-inequality` (board score 1.5028503020710076; retrieved 2026-07-04). The board scores a vector by the floating-point semantics

$$\max_m (\text{conv}(f, f)_m \cdot h) / \left(\sum_i f_i \cdot h \right)^2, \quad h = \frac{1}{2n},$$

which by Lemma 1 is exactly the continuum functional $\sup_t g / (\int f)^2$ of the associated step function, up to the board’s float rounding — which we now remove.

Theorem 2 (first autocorrelation inequality; construction: Hyra [5]).

$$C_1 \leq Q_1 := \frac{1661053907432921000898809384050165075655877436050568807950600634368000}{1105269037870172510060979851604850227379926078898766312058428453057449},$$

and

$$1.5028503020710078765191887 \leq Q_1 < 1.5028503020710078765191888.$$

In particular $C_1 \leq 1.5028503020710078765191888 < \pi/2$.

Proof. Each entry of the retrieved vector is an IEEE-754 double, hence an exact dyadic rational; over the common denominator $D = 2^{117}$ the integer numerators A_i have at most 110 bits, all are nonnegative (exact check, zero violations), and $S := \sum_i A_i$ has 119 bits, with $S > 0$. The step function $F = \sum_i (A_i/D) \mathbf{1}_{I_i}$ is admissible for (C_1) after normalizing by $\int F$; since the functional $\sup_t (F * F) / (\int F)^2$ is invariant under positive scaling, no explicit renormalization is needed. All $2n - 1 = 179999$ integer autoconvolution coefficients $c_m = \sum_{i+j=m} A_i A_j$ were computed by Kronecker substitution with byte-aligned slot width $b = 248$ bits (the checksum $\sum_m c_m = S^2$ holds exactly); the maximum $M = \max_m c_m$ has 221 bits and occurs at $m^* = 116347$, i.e. at $t^* = -\frac{1}{2} + (m^* + 1)h = 6587/45000 \approx 0.14638$. By Lemma 1(iii) and the cancellation of D and h ,

$$\frac{\sup_t (F * F)(t)}{(\int F)^2} = \frac{2n M}{S^2} = Q_1$$

after reduction to lowest terms (70-digit numerator and denominator), and $C_1 \leq Q_1$ because every admissible f upper-bounds the infimum. The decimal enclosure is obtained by exact integer division of $10^{25} \text{num}(Q_1)$ by $\text{den}(Q_1)$ (floor and ceiling). The computation runs in 189s with no floating-point operation on the certified path (script `exact_board2.py`, Appendix A). $\square$

Remark 3 (relation to the board value). $\text{float}(Q_1) = 1.5028503020710078$ agrees with the reported board score 1.5028503020710076 to one unit in the last place (2.2×10^{-16} ; the exact value is a hair *above* the float score, which is a rounding of the same vector). An FFT-based float re-verification of the vector reproduces the board score to 4.4×10^{-16} . The float agreement plays no role in the proof; it is a sanity cross-check. A literal rerun of the board’s own $O(n^2)$ convolution code path is pending (Section 7).

Remark 4 (landscape and scope of the claim). To our knowledge, Theorem 2 is the first exact-arithmetic-certified upper bound on C_1 in the sub-1.51 regime; we deliberately do *not* describe it as the first proven sub-1.51 bound simpliciter without the exact-arithmetic qualifier, since the boundary between “numerical construction” and “proof” is drawn differently by different authors, and the constructions below are genuine mathematical objects whose printed evaluations

are merely uncertified. The comparison landscape: the analytic bound $\pi/2$ [12] is fully proven; Matolcsi–Vinuesa’s 1.51237... (LOQO) and 1.50972... ($n = 208$ LP fix-point, stated as $C_1 \leq 1.5098$) rest on floating-point evaluations of printed 8-digit data [9]; the AI-search reports 1.5068 [13], 1.5053 and 1.5032 [10, 4], 1.50286 [15], 1.502862 (Together AI, tabulated in [14]) and 1.503871 [14] are floating point, and all lie *above* Q_1 . Theorem 2 says nothing about the lower side: the peer-reviewed $C_1 \geq 1.28$ [3] and the interval-certified $C_1 \geq 1.2937$ of the preprint [7] stand as published, and the gap $[1.2937, Q_1]$ remains wide open.

Remark 5 (citation hygiene: 1.50992). The digit string 1.50992, widely quoted as the Matolcsi–Vinuesa record, does not appear in [9]: that paper prints $\|f * f\|_\infty = 1.50972\dots$ for its best construction and states the bound as 1.5098. The string 1.50992 enters the literature through Cloninger–Steinerberger, who describe the state of the art as “ $1.2749 \leq c \leq 1.50992$ ” [3, p. 2]. Quotations of 1.50992 should therefore cite [3], not [9].

4 The second constant: a certified lower bound, and two published vectors re-certified

The candidate is a vector with $n = 524288$ cells, found by Hyra [5]: the current leader of the board **second-autocorrelation-inequality** (board score 0.9629011010961752; retrieved 2026-07-04). Its entries happen to be nonnegative *integers* (at most 31 bits), which simplifies the exact path: no denominator clearing is needed.

Theorem 6 (second autocorrelation inequality; construction: Hyra [5]).

$$C_2 \geq Q_2 := \frac{140651861665566489683881393353250795846281833}{14607093242021125986978346843833325818535926},$$

and

$$0.9629011010961756912885013 < Q_2 < 0.9629011010961756912885014.$$

Proof. Let $A_0, \dots, A_{n-1}$ be the integer heights and c_m their exact autoconvolution, computed by Kronecker substitution with slot width $b = 96$ bits (the packed integer has $\approx 5.03 \times 10^7$ bits; the big-integer squaring took 553 s); the checksum $\sum_m c_m = (\sum_i A_i)^2$ holds exactly. Write

$$S_1 = \sum_m c_m = 7973480668564803823766721, \quad \Lambda = \max_m c_m = 12213062984825962404,$$

$$S_2 = \sum_{m=-1}^{2n-2} (c_m^2 + c_m c_{m+1} + c_{m+1}^2) = 281303723331132979367762786706501591692563666,$$

with zero padding as in Lemma 1(iv). By parts (iii) and (iv) of the lemma, for the associated step function F (any positive scale, any grid width — both cancel),

$$\frac{\|F * F\|_2^2}{\|F * F\|_1 \|F * F\|_\infty} = \frac{h^3 S_2 / 3}{(h^2 S_1)(h\Lambda)} = \frac{S_2}{3 S_1 \Lambda} = Q_2$$

in lowest terms, and $C_2 \geq Q_2$ since the supremum dominates the value at F . The decimal enclosure is exact integer division. The total certified computation took 576 s (script `exact_board3.py`, Appendix A); the board’s own float verifier, re-run on the vector, reproduces the reported score bit-for-bit, and $\text{float}(Q_2) = 0.9629011010961757$ differs from it by 4.4×10^{-16} (rounding inside the verifier’s FFT path). $\square$

We also certify, by the same lemma, the two strongest published C_2 construction vectors.

Proposition 7 (re-certification of two published C_2 vectors). (a) Let v_{JJ} be the 2540-entry coefficient vector `function.py` published in the repository accompanying Jaech–Joseph [6]. Its exact continuum ratio lies in

$$[0.9413628949737643225960591, 0.9413628949737643225960592]$$

(directed 25-digit enclosure; the exact value begins 0.9413628949737643225960591212...); in particular the Jaech–Joseph claim $C_2 \geq 0.94136$, which in [6] rests on Simpson/Riemann quadrature, is now certified. Q_2 exceeds this value by more than 0.0215382061224113.

(b) Let v_{BL} be the 575 printed integers of Boyer–Li [2]. Our independent exact evaluation reproduces their published fraction

$$\frac{96086089410408840289339769}{106577013451431545242354944} = 0.9015648524829085404822199\dots$$

identically — the two exact computations agree to the last digit. Their in-paper evaluation is fully rigorous.

Proof. Part (a): the file entries are parsed as IEEE-754 doubles (exact dyadic rationals; 2540 entries, all nonnegative), scaled to integers over the common power-of-two denominator, and evaluated by the integer formula $S_2/(3S_1\Lambda)$ exactly as in Theorem 6 (script `certify.py` and enclosure script, Appendix A). Part (b): the 575 integers are used directly. Both computations run in seconds. $\square$

Remark 8 (margins and the state of the C_2 problem). The headline comparison is proven-over-proven: Q_2 exceeds the strongest previously certifiable value — the Jaech–Joseph vector, certified in Proposition 7(a) — by $+0.02153820612241136\dots$ We do not headline the larger gap (≈ 0.0613) over the Boyer–Li fraction, because the Jaech–Joseph float claim was genuine and is now certified. Float-only reports larger than 0.94136 exist — 0.961 [4], 0.961206 (Together AI, tabulated in [14]), 0.962694 [14] — and all lie below Q_2 . On the upper side we know nothing beyond Hölder’s trivial $C_2 \leq 1$; whether $C_2 < 1$ is open. Footnote of record: the abstract and introduction of [6] quote the Boyer–Li value as .901562; the exact value is 0.9015648... as above.

5 The third constant: the signed variant certified

The candidate is a *signed* vector with $n = 100000$ cells, found by the agent OrganonAgent [11]: the current leader (board score 1.4523043331831582; retrieved 2026-07-04) of the board `third-autocorrelation-inequality`.

Theorem 9 (third autocorrelation inequality, signed; construction: OrganonAgent [11]).

$$C_3 \leq Q_3 := \frac{11753128449293701953238517385067272445617294540800000}{8092744874989952471246071559466128309374865340943729},$$

and

$$1.452304333183157960701884 < Q_3 < 1.452304333183157960701885.$$

Proof. The entries are IEEE-754 doubles; over the common denominator $D = 2^{68}$ the signed integer numerators A_i have at most 77 bits and $S := \sum_i A_i$ has 87 bits, $S \neq 0$. Splitting $A = P - Q$ with $P, Q \geq 0$, the three nonnegative autoconvolutions c_{PP}, c_{PQ}, c_{QQ} were computed by Kronecker substitution with 22-byte slots and combined as $c = c_{PP} - 2c_{PQ} + c_{QQ}$; eight random positions were re-verified against direct $O(n)$ integer sums, and the checksum $\sum_m c_m = S^2$ holds exactly. The maximum $M = \max_m c_m$ is positive (so the absolute value in the board functional is inert — as it must be, by Remark 13) and occurs at $m^* = 163479$, i.e. at $t^* = -\frac{1}{2} + (m^* + 1)h = 1587/5000 = 0.3174$ exactly. By Lemma 1(iii), with D, h cancelling,

$$\frac{\max_t (F * F)(t)}{(\int F)^2} = \frac{2n M}{S^2} = Q_3$$

in lowest terms, and $C_3 \leq Q_3$. The decimal enclosure is exact integer division; the certified computation took 55 s (script `exact_board4.py`, Appendix A). $\text{float}(Q_3) = 1.452304333183158$ agrees with the reported board score to a relative 1.5×10^{-16} (here the exact value sits a hair *below* the reported float — again a rounding of the same vector, not an improvement); the float cross-check used FFT convolution, and a literal rerun of the board’s $O(n^2)$ path is pending (Section 7). $\square$

To our knowledge — subject to the arXiv sweep listed in Section 7 — Theorem 9 is the first rigorously certified nontrivial upper bound for the signed variant. (The qualifier *nontrivial* excludes the inherited bound $C_3 \leq C_1 \leq \pi/2$, valid because every nonnegative admissible f is also signed-admissible and the two constants are infima of the same functional.) The two classical reference points for this constant are float reports; we certify both.

Proposition 10 (re-certification of two published C_3 vectors). *(a) Let v_{MV} be the $n = 150$ signed step vector printed in the appendix of Matolcsi–Vinuesa [9] (coefficients parsed as exact decimals from the arXiv source). Then, exactly,*

$$\begin{aligned} \frac{\max_t (v * v)(t)}{(\int v)^2} &= \frac{174972681773148828}{119999999644045921} \\ &\in [1.458105685768062453851950, 1.458105685768062453851951], \end{aligned}$$

confirming the printed value “1.45810...”. However,

$$\frac{\|v * v\|_\infty}{(\int v)^2} \in [2.426895802097483067204798, 2.426895802097483067204799],$$

attained at a negative trough; see Remark 11.

(b) Let v_{AE} be the public $n = 400$ AlphaEvolve vector (`height_sequence_3` of the results notebook [1]; reported as 1.4557 in [10, 4]). Then, exactly,

$$\begin{aligned} \frac{\max_t (v * v)(t)}{(\int v)^2} &= \frac{116451423625336345095200}{7999999996821408075961} \\ &\in [1.455642795374540494110788, 1.455642795374540494110789], \end{aligned}$$

*while $\|v * v\|_\infty / (\int v)^2 = 4.334046524387984\dots$, again attained on the negative side.*

Consequently Q_3 lies below the certified AlphaEvolve value by more than $3.3384621913825 \times 10^{-3}$ and below the certified Matolcsi–Vinuesa value by more than $5.8013525849044 \times 10^{-3}$.

Proof. Direct $O(n^2)$ exact integer autoconvolution after exact decimal (resp. dyadic) parsing, as in Section 2.1; both runs take seconds (script `cert_published_c3.py` and enclosure script, Appendix A). The margins are exact rational subtractions with directed decimal enclosures. $\square$

Remark 11 (a correction of record in [9]). The closing remark of [9] presents the $n = 150$ signed example with the statement “ $\|f * f\|_\infty = 1.45810\dots$ ”. Taken literally (as an L^∞ norm) this is incorrect for the printed vector: Proposition 10(a) shows its ∞ -norm ratio is $2.42689\dots$, and $1.45810\dots$ is the *signed maximum* $\max_t (f * f)(t)$. The example is thus an example for the signed-maximum constant C_3 — which is how [4, Problem 6.4(b)] files it — and certifies (now exactly) $C_3 \leq 1.45810568576806245\dots$; it says nothing about the ∞ -norm variant. The same caution applies to the AlphaEvolve vector (∞ -norm ratio $4.334\dots$) and to the Theorem 9 leader itself (see Figure 1).

Remark 12 (disambiguation: the $\|\cdot\|_\infty$ cousin is a different problem). Problem 6.4 of [4] defines *two* constants for signed f : $C_{6.4}$, with the absolute value *inside* the maximum ($\max_t |(f * f)(t)| = \|f * f\|_\infty$), and $C'_{6.4}$, with the absolute value *outside* ($|\max_t (f * f)(t)|$) — the constant C_3 of this note. The reported values for the cousin $C_{6.4}$ are 1.4993 , improved by AlphaEvolve to 1.4688 [4]; these must never be compared with values for C_3 . The confusion is live in the literature: the ∞ -norm phrasing of [9] discussed in Remark 11 is its seed, and, e.g., the task specification for this problem in [14] prints the formula with the absolute value inside the maximum while quoting the signed-maximum record 1.4556427953745406 as the target — two functionals that differ by a factor of about 3 on the record vectors themselves. Every value in this section is for C_3 (absolute value outside), the semantics of the arena board and of [4, 6.4(b)].

Remark 13 (the lower side of C_3). Since f is supported on $[-\frac{1}{4}, \frac{1}{4}]$, $g = f * f$ is supported on $[-\frac{1}{2}, \frac{1}{2}]$, an interval of length 1, and $\int g = (\int f)^2$ by Fubini; hence $\max_t g \geq \int g = (\int f)^2 > 0$, giving the trivial bound $C_3 \geq 1$ (and showing the absolute value in the definition is inert). Nothing better is known to us: the nonnegative-case lower bounds $C_1 \geq 1.28$ [3] and $C_1 \geq 1.2937$ [7] do *not* apply to signed f — their proofs use $f \geq 0$ throughout — and no signed analogue has been published to our knowledge. The proven bracket for the signed constant after this note is therefore $1 \leq C_3 \leq 1.452304333183157960701885$, with all of the lower side open.

6 Summary: what is certified, and what is not

Table 1 collects the three certified statements; Table 2 collects the four re-certifications. Directions matter and are stated explicitly: a construction certifies an upper bound for an infimum (C_1 , C_3) or a lower bound for a supremum (C_2), and says nothing about the opposite side.

constant	certified here (exact)	direction	opposite side (quoted)	agent
C_1	$Q_1 = 1.50285030207100788\dots$	upper	≥ 1.28 [3]; ≥ 1.2937 [7]	Hyra
C_2	$Q_2 = 0.96290110109617569\dots$	lower	≤ 1 (Hölder; < 1 open)	Hyra
C_3	$Q_3 = 1.45230433318315796\dots$	upper	≥ 1 (trivial; nothing better)	OrganonAgent

Table 1: The three certified bounds (Theorems 2, 6 and 9). Each exact value is a ratio of explicit integers printed in the theorem statements; the decimals shown here are truncations of the certified enclosures.

Three cautions, stated bluntly so that this note is not over-read. First, the certificates evaluate the existing arena leader vectors; the constructions are the agents’ [5, 11], and the certified values

vector	problem	n	published claim	exact value (this note)	verdict
Boyer–Li [2]	C_2	575	0.90156485... (exact)	identical fraction	confirmed
Jaech–Joseph [6]	C_2	2540	≥ 0.94136 (float)	0.9413628949737643...	now certified
Matolcsi–Vinuesa [9]	C_3	150	$\ f * f\ _\infty = 1.45810...$ (float)	$\max_t$: 1.4581056857... $\ \cdot\ _\infty$: 2.4268958020...	corrected (Rem. 11)
AlphaEvolve [1]	C_3	400	1.4557 (float)	1.4556427953745404...	now certified

Table 2: Exact re-certifications of published construction vectors (Propositions 7 and 10).

agree with the boards’ float scores to about one unit in the last place — nothing on any leaderboard moves because of this note. Second, exactness is not optimality: Q_1, Q_2, Q_3 are values of particular step functions, and there is no reason to believe any of them is close to the true constant on the open side ($C_1 \in [1.2937, Q_1]$, $C_2 \in [Q_2, 1]$, $C_3 \in [1, Q_3]$ are the honest brackets). Third, the “first certified” claims in Remark 4 and Section 5 are qualified by “to our knowledge” and by the pending sweep of Section 7; they are claims about certification, not about the underlying mathematics, whose credit lies with the works cited.

7 Pending-verification ledger

For full transparency, items *not* machine-verified at the time of writing, none of which affects the exact evaluations of Theorems 2, 6 and 9:

1. **Literal verifier reruns.** The floating-point cross-checks of the Theorem 2 and Theorem 9 vectors used FFT-equivalent convolution (agreement with the reported board scores $\leq 4.5 \times 10^{-16}$; the exact integer argmax matched the FFT argmax in both cases). A byte-exact rerun of the boards’ literal $O(n^2)$ `np.convolve` code path on these vectors is pending (scheduled off-peak). This affects only the sanity cross-checks, not the proofs.
2. **Together AI values.** The values 1.502862, 0.961206, 1.454555 attributed to a Together AI team are known to us only as float entries in the comparison table of [14]; their certification status, and the exact provenance of the corresponding vectors, are unverified. Our exact bounds are on the favorable side of all three, so nothing here depends on them.
3. **Same-day literature sweep for C_3 .** The “first rigorously certified upper bound for the signed variant” claim should be preceded, at submission time, by a same-day arXiv sweep for the signed constant; the claim is stated with that qualification.
4. **Provenance freezes.** The Jaech–Joseph vector was taken from their public repository [6] and the AlphaEvolve vector from the public results notebook [1]; the three arena leader vectors were retrieved from the live boards on 2026-07-04. Cryptographic hashes of the exact copies certified here are to be frozen into the ancillary artifact bundle. (We note in passing that [6] describes its high-resolution function as a 2399-interval object while the repository coefficient file contains 2540 entries; our certificate applies to the repository file as retrieved.)

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computational papers and artifacts published carefully enough to be independently re-evaluated, certified, and built upon. This note was prepared with CHRONOS, ProjectForty2’s autonomous research agent.

A Reproducibility

All scripts, input vectors, and verifier outputs accompany this note as ancillary files, in the directories `scripts/`, `vectors/`, and `certs/`; every reproduction command in the accompanying README was re-verified end-to-end from this bundle. Cryptographic hashes are to be frozen per Section 7, item 4.

Headline bounds (exact; certify Theorems 2, 6 and 9).

- `scripts/exact_board2.py` — Theorem 2. Input: the $n = 90000$ Hyra vector as retrieved. Output must reproduce `certs/exact_board2_out.txt`: the exact fraction Q_1 , the 25-digit enclosure, $\text{argmax } m^* = 116347$, slot width, checksum, and the nonnegativity audit. Runtime 189 s.
- `scripts/exact_board3.py` — Theorem 6. Input: the $n = 524288$ integer Hyra vector. Output must reproduce `certs/exact_board3_out.txt`: S_1 , Λ , S_2 , the exact fraction Q_2 , and the checksum PASS. Runtime 576 s (big-integer squaring 553 s).
- `scripts/exact_board4.py` — Theorem 9. Input: the $n = 100000$ signed OrganonAgent vector. Output must reproduce `certs/exact_board4_out.txt`: the exact fraction Q_3 , the enclosure, $\text{argmax } m^* = 163479$, the signed Kronecker split, and the spot-check line. Runtime 55 s.

Published-vector re-certifications (Propositions 7 and 10).

- `scripts/certify.py` — exact evaluation of the Boyer–Li 575-integer vector and the Jaech–Joseph repository vector (`coeffBL.txt`, `jj_function.py`); reproduces the Boyer–Li fraction identically and the Jaech–Joseph exact ratio.
- `scripts/cert_published_c3.py` — exact signed evaluation of the Matolcsi–Vinuesa $n = 150$ printed vector (decimal-parsed from the arXiv source of [9]; `mv150_printed.txt`) and the AlphaEvolve $n = 400$ public vector, including the ∞ -norm ratios of Remark 11. Output: `certs/cert_published_c3_out.txt`.
- `scripts/recert_enclosures.py` — the “enclosure script” cited in the proofs of Propositions 7 and 10: it computes the directed decimal enclosures of all re-certified values and margins by exact `Fraction` floor/ceil. Output: `certs/recert_enclosures.txt`, whose decimals byte-match the values quoted in the text.

Figure. `make_figure.py` regenerates Figure 1 from the envelope-downsampled display data (`fig_data_ac.json`); the figure is display-only and carries no certified content.

Machine environment. An Apple-silicon Mac (published-vector re-certifications, source extraction, figure and note build) and a workstation-class NVIDIA GPU host (the three large certifications; the $n = 524288$ evaluation squares a ≈ 50 -million-bit integer in 553s). All certified arithmetic is CPython arbitrary-precision `int`/`Fraction`; no GPU arithmetic touches any certified path.

References

- [1] Google DeepMind, *AlphaEvolve results repository* (`mathematical_results.ipynb`, variable `height_sequence_3`), https://github.com/google-deepmind/alphaevolve_results, retrieved 2026-07-04.
- [2] C. Boyer and Z. K. Li, *An improved example for an autoconvolution inequality*, arXiv:2506.16750 [math.CA], 2025.
- [3] A. Cloninger and S. Steinerberger, *On suprema of autoconvolutions with an application to Sidon sets*, Proc. Amer. Math. Soc. **145** (2017), 3191–3200; arXiv:1403.7988.
- [4] B. Georgiev, J. Gómez-Serrano, T. Tao, and A. Z. Wagner, *Mathematical exploration and discovery at scale*, arXiv:2511.02864, 2025.
- [5] Hyra (pseudonymous AI search agent), step-function constructions for the first and second autocorrelation inequalities ($n = 90000$, board score 1.5028503020710076; $n = 524288$, board score 0.9629011010961752), EinsteinArena, problems `first-autocorrelation-inequality` and `second-autocorrelation-inequality`, <https://einsteinarena.com/problems/first-autocorrelation-inequality>, retrieved 2026-07-04.
- [6] A. Jaech and A. Joseph, *Further improvements to the lower bound for an autoconvolution inequality*, arXiv:2508.02803 [math.CA], 2025. Coefficient file: https://github.com/ajaech/h/autocorrelation_inequality.
- [7] S. Kim and M. Pilanci, *AI-assisted discovery of convex relaxations via dual agents*, arXiv:2606.31182, 2026.
- [8] G. Martin and K. O’Bryant, *The supremum of autoconvolutions, with applications to additive number theory*, Illinois J. Math. **53** (2009), no. 1, 219–235.
- [9] M. Matolcsi and C. Vinuesa, *Improved bounds on the supremum of autoconvolutions*, J. Math. Anal. Appl. **372** (2010), no. 2, 439–447; arXiv:0907.1379. Printed coefficient lists cited here follow the arXiv source.
- [10] A. Novikov et al., *AlphaEvolve: a coding agent for scientific and algorithmic discovery*, arXiv:2506.13131, 2025.
- [11] OrganonAgent (pseudonymous AI search agent), signed step-function construction for the third autocorrelation inequality ($n = 100000$, board score 1.4523043331831582), EinsteinArena, problem `third-autocorrelation-inequality`, <https://einsteinarena.com/problems/third-autocorrelation-inequality>, retrieved 2026-07-04.
- [12] A. Schinzel and W. M. Schmidt, *Comparison of L^1 - and L^∞ -norms of squares of polynomials*, Acta Arith. **104** (2002), no. 3, 283–296.

- [13] Y. Wang et al., *ThetaEvolve: test-time learning on open problems*, arXiv:2511.23473, 2025.
- [14] H. Ye et al., *Evaluation-driven scaling for scientific discovery*, arXiv:2604.19341, 2026. (SimpleTES.)
- [15] M. Yuksekgonul et al., *Learning to discover at test time*, arXiv:2601.16175, 2026. (TTT-Discover.)